

USART, a byte-oriented two-wire serial interface, a programmable watchdog timer with internal oscillator, an SPI serial port and six software-selectable power-saving modes.

Piezobuzzer PZ1 is used as an audible indicator for true, fake, random touches and system halt. It is controlled from port pin PD6 of the microcontroller with the help of transistor T1. The iButton socket is connected to port pin PD2. Pull-up resistor R1 is used as required from the 1-wire protocol. Port pin PD4 controls the relay operation through optocoupler PC817 (IC2). The door locking mechanism, say, for a slide door, is connected to the contacts of relay RL1, which closes after some time automatically.

Whenever someone attempts to gain access by touching the iButton

(IC5) using his own iButton, the firmware inside the AVR reads its unique ID and matches with the ID in the firmware. If the ID matches, port pin PD4 goes high, the internal LED of the optocoupler (IC2) glows and relay RL1 energises for the predefined time. Simultaneously, the buzzer sounds to indicate grant of access.

Thereafter, relay RL1 de-energises. The buzzer gets the PWM signal to produce sound from port pin PD6 of the microcontroller. The 1-wire communication is done through interrupt port pin PD2 (INT0), so the AVR does not have to poll the pin but respond when any change in signal is detected on the interrupt pin. Switch S1 is used for manual reset.

Port pin PD1 of the microcontroller is used to interface with the hyper terminal of the PC through RS-232 interface MAX232 IC (IC3) for iButton verification and checking. The microcon-

troller provides a transmit channel for serial data transfer. Transmit data pin (TXD) is specified at port pin PD1. The microcontroller is connected to T1 IN (pin 11) of MAX232. T1 OUT (pin 14) of IC3 is connected to pin 2 of the COM port connector. The signals provided on these pins are TTL-level and must be boosted and inverted through a MAX232 converter to comply with the RS-232 standard.

The MAX232 has two internal charge pumps that convert +5V into $\pm 10V$ (unloaded) for RS-232 driver operation. The first converter uses capacitor C6 to double the +5V input to +10V on capacitor C8 at pin 2. The second converter uses capacitor C5 to invert +10V to -10V on capacitor C4 at pin 6.

The power supply for this circuit is derived from 230V, 50Hz AC mains. Transformer X1 steps down 230V, 50Hz AC mains to deliver a secondary output of 9V, 300 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7806 (IC4). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R5 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for the secure digital access system is shown in Fig. 5 and its component layout in Fig. 6. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Connect the assembled circuit to the COM port of the computer. The iButton access code and message is transferred to the PC through the Hyper Terminal

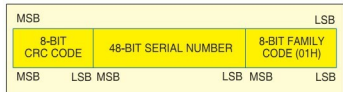


Fig. 3: 64-bit lasered ROM

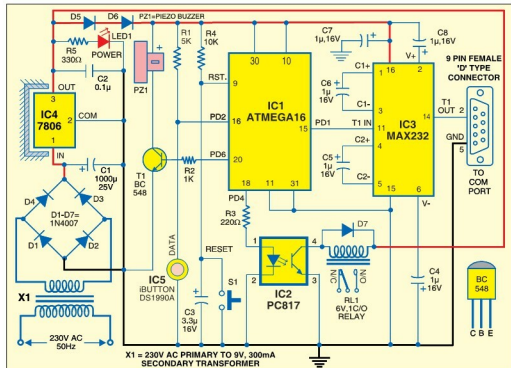


Fig. 4: Circuit of the secure digital access system using iButton